

9 Disproof

Each of the following statements is either true or false. If a statement is true, prove it. If a statement is false, disprove it. These exercises are cumulative, covering all topics addressed in Chapters 1-9.

2. For every natural number n , the integer $2n^2 - 4n + 31$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 31$. Then $2n^2 - 4n + 31 = 2(31^2) - 4(31) + 31 = 1829 = 31 \cdot 59$, which is not prime. $\square$

4. For every natural number n , the integer $n^2 + 17n + 17$ is prime.

Disproof. The statement is false. For a counterexample, let $n = 1$. Then $n^2 + 17n + 17 = 1^2 + 17(1) + 17 = 35 = 7 \cdot 5$, which is not prime. $\square$

6. If A, B, C and D are sets, then $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Proof. Observe the following sequence of equalities.

$$\begin{aligned}
 (A \times B) \cap (C \times D) &= \{(x, y) : (x, y) \in A \times B \wedge (x, y) \in C \times D\} && \text{(definition of } \cap \text{)} \\
 &= \{(x, y) : (x \in A \wedge y \in B) \wedge (x \in C \wedge y \in D)\} && \text{(definition of } \times \text{)} \\
 &= \{(x, y) : (x \in A \wedge x \in C) \wedge (y \in B \wedge y \in D)\} && \text{(regrouping)} \\
 &= \{(x, y) : x \in A \cap C \wedge y \in B \cap D\} && \text{(definition of } \cap \text{)} \\
 &= \{(x, y) : (x, y) \in (A \cap C) \times (B \cap D)\} && \text{(definition of } \times \text{)} \\
 &= (A \cap C) \times (B \cap D) && \text{(definition of set-builder)}
 \end{aligned}$$

$\square$

8. If A, B and C are sets, then $A - (B \cup C) = (A - B) \cup (A - C)$.

Disproof. The statement is false. For a counterexample, let $A = \{1, 2\}$, $B = \{1\}$ and $C = \{2\}$. Then $A - (B \cup C) = \{1, 2\} - \{1, 2\} = \emptyset$, but $(A - B) \cup (A - C) = \{2\} \cup \{1\} = \{1, 2\}$. Since $\emptyset \neq \{1, 2\}$, the two sides are not equal. $\square$

10. If A and B are sets and $A \cap B = \emptyset$, then $\mathcal{P}(A) - \mathcal{P}(B) \subseteq \mathcal{P}(A - B)$.

Proof. Suppose $A \cap B = \emptyset$. Let $X \in \mathcal{P}(A) - \mathcal{P}(B)$. Then $X \in \mathcal{P}(A)$ and $X \notin \mathcal{P}(B)$, so $X \subseteq A$. Now we claim $A - B = A$. Let $a \in A - B$. Then $a \in A$ and $a \notin B$. Thus every element of $A - B$ is an element of A , so $A - B \subseteq A$. Conversely, let $a \in A$. If $a \in B$, then $a \in A$ and $a \in B$, so $a \in A \cap B = \emptyset$, which is impossible. Thus $a \notin B$. So we have $a \in A$ and $a \notin B$, which means $a \in A - B$. Hence $A \subseteq A - B$. Therefore $A - B = A$. So we have $X \subseteq A = A - B$, thus $X \in \mathcal{P}(A - B)$. Since X was arbitrary, $\mathcal{P}(A) - \mathcal{P}(B) \subseteq \mathcal{P}(A - B)$. $\square$

12. If $a, b, c \in \mathbb{N}$ and ab, bc and ac all have the same parity, then a, b and c all have the same parity.

Disproof. The statement is false. For a counterexample, let $a = 2$, $b = 1$ and $c = 2$. Then $ab = 2$, $bc = 2$ and $ac = 4$ are all even, so ab, bc and ac have the same parity. But a and c are even while b is odd, so a, b and c do not all have the same parity. $\square$

14. If A and B are sets, then $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.

Proof. First we prove $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$. Suppose $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. By definition of intersection, this means $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$. By definition of power set, $X \subseteq A$ and $X \subseteq B$. Since $X \subseteq A$ and $X \subseteq B$, every element of X lies in both A and B , hence in $A \cap B$; therefore $X \subseteq A \cap B$. By definition of power set, this means $X \in \mathcal{P}(A \cap B)$. Since X was arbitrary, $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$.

Now we show $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$. Suppose $X \in \mathcal{P}(A \cap B)$. By definition of power set, this means $X \subseteq A \cap B$; in other words, every element of X lies in both A and B . Thus $X \subseteq A$ and $X \subseteq B$. Hence, by definition of power set, $X \in \mathcal{P}(A)$ and $X \in \mathcal{P}(B)$. Thus, by definition of intersection, $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$. Since X was arbitrary, $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$.

Therefore, since $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$ and $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$, it follows that $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$. $\square$

16. If A and B are finite sets, then $|A \cup B| = |A| + |B|$.

Disproof. The statement is false. For a counterexample, let $A = \{1, 2\}$ and $B = \{1\}$. Then $|A \cup B| = |\{1, 2\}| = 2$, and $|A| + |B| = 2 + 1 = 3$. Since $2 \neq 3$, we have $|A \cup B| \neq |A| + |B|$. $\square$

18. If $a, b, c \in \mathbb{N}$, then at least one of $a - b$, $a + c$, and $b - c$ is even.

Proof. Assume $a, b, c \in \mathbb{N}$. Then each is either even or odd. By the pigeonhole principle, at least two must have the same parity. If a and b have the same parity, then $a - b$ is even. If a and c have the same parity, then $a + c$ is even. If b and c have the same parity, then $b - c$ is even. Therefore at least one of $a - b, a + c$ and $b - c$ is even. $\square$

20. There exist prime numbers p and q for which $p - q = 1000$.

Proof. Let p be the prime number 2009 and let q be the prime number 1009. Then $p - q = 2009 - 1009 = 1000$. $\square$